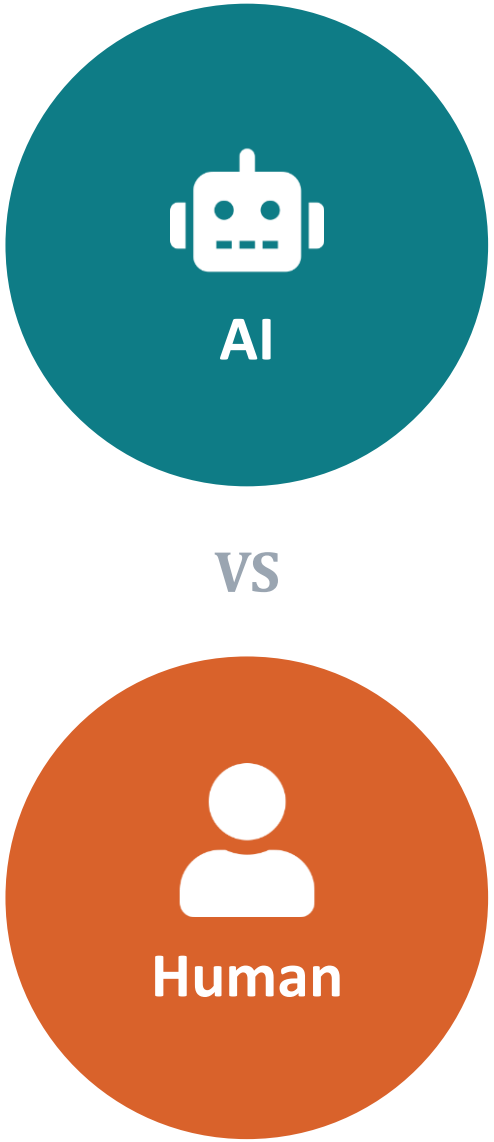
PROBABILITY AND STATISTICS SEMESTER PROJECT REVIEW 1

A statistical comparison of human written and AI-generated code across programming languages



A study of code length and comment characteristics.

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Intro and problem statement

Background:

Generative AI tools such as ChatGPT, Claude, Gemini and such write programming code on request and AI assisted programming is spreading fast, including in coding heavy courses, and even in the job sector, with some companies replacing junior engineers with AI models.

Why all of this matters:

Understanding how AI generated code differs statistically from human-written code supports research on code provenance and responsible AI use.

Scope:

This project does not detect misconduct or prove authorship. It measures and compares code characteristics using statistical methods.

OUR RESEARCH PROBLEM:

Are there statistically significant differences in code characteristics between human written and AI generated code and how do these differences vary?

Measured: line count, character count, approximate comment lines

Languages analyzed: Java, Python, C, C++

Objectives and research questions

Objectives

1. To compare the line counts and the character counts of human written + AI generated code.
2. Compare approximate comment line counts.
3. To investigate whether the origin (Human or AI written) of the code is associated with programming language.
4. To apply descriptive statistics and inferential tests (hypothesis tests and confidence intervals) to our dataset.

Research questions

- **RQ1** Do both groups show difference in the mean line count and character count?
- **RQ2** Do their comment line characteristics differ?
- **RQ3** Is the code origin associated with the programming language it's written in?
- **RQ4** Do the observed differences vary across the languages: Java, Python, C and C++?

Literature review

sno	Authors and year:	Title:	Journal or conference:	What it does:	Relevance:	Limitation:
1	Boukili, El Garouani & Riffi (2026)	A dataset for human-written and AI-generated code source classification	Data in Brief, 65, 112527	It releases 10000 labelled samples where 5000 are human 5000 are from ChatGPT in java, python, C and C++, including XGBoost and LSTM classifiers	It is the main thing, the source of the dataset analysed in this project.	Human code (CodeNet) and AI code (ChatGPT) come from different collection places.
2	Demirok, Kutlu & Mergen (2026)	MultiAIGCD: A comprehensive dataset for AI generated code detection covering multiple languages, models, prompts, and scenarios	Neural Computing and Applications, 38(16), 687	Large dataset in python, java and go from six LLMs and three scenarios, it basically benchmarks detection models.	Detection performance varies by language, so that means each language should be analysed separately.	It's a machine learning benchmark, not a statistical analysis of code measures.
3	Gurioli, Gabbrielli & Zacchiroli (2025)	Is This You, LLM? Recognizing AI-written Programs with Multilingual Code Stylometry	IEEE SANER 2025 (arXiv:2412.14611)	Transformer based classifier for AI code stylometry across 10 languages, and it releases the H-AIRosettaMP dataset which holds 121247 snippets.	Shows that style differences between human and AI code can be studied across languages.	Focuses on classification here average accuracy 84.1%, not really hypothesis testing.
4	Demirok & Kutlu (2024)	AIGCodeSet: A new annotated dataset for AI generated code detection	arXiv preprint (arXiv:2412.16594)	2828 AI-generated and 4755 human-written python programs; AI code from CodeLlama, Codestral and Gemini 1.5 flash, baseline detectors.	AI code comes from many different LLMs, so our conclusions stay specific to our dataset.	python only, detection is the focus here.
5	Li, Treude & Turel (2026)	Do comments and expertise still matter? An experiment on programmers' adoption of AI-generated JavaScript code	Journal of Systems and Software, 231, 112634	It's an experiment with 173 programmers: comments in AI generated code significantly increased its adoption, regardless of expertise.	Our motivation to studying code comments as a code characteristic.	Studies adoption behaviour, not exactly that if comment counts identify AI code.

Papers 1 to 4 build datasets and detectors, paper 5 had us look into comments. Our project is a focused statistical comparison. Paper 1 (a dataset for human written and Ai generated code source classification is the dataset we used for this project).

Motivation for this study

Why it this important

AI code generators, they make it really easy to submit programming work without independent effort. Coding-intensive institutions need evidence-based understanding of how AI code differs from human code. Also, as it is being increasingly used in the workplace, character/number of lines can be used as a factor to analyze code efficiency

What the study does

In this phase, our study compares human-written and AI generated code on line count, character count and approximate comment lines, using confidence intervals and hypothesis tests.

What it can be used for

Preliminary evidence for researchers and developers building code provenance and academic integrity tools, and a statistical baseline for later work.

What it does not do

It cannot prove a given program is AI written or that misconduct occurred. A human program can resemble AI code and the vice-versa.

Basically, earlier work built datasets and machine learning detectors. Our project is a focused statistical comparison that interprets effect sizes and uncertainty, language by language.

Dataset and variables

DATASET:
A dataset for human-written and AI-generated code source classification (Boukili et al., 2026)

SOURCE:
Mendeley Data

SAMPLE SIZE:
10000 code samples

GROUPS:
5000 AI-generated and 5000 human-written (human: CodeNet; AI: ChatGPT)

LANGUAGES:
java, python, C and C++

DATA FIELDS (for this review):
Problem ID, source code, origin, language and source

Variable	Meaning	Kind
Generated	AI or Human	Original
Language	Programming language	Original
Line_Count	Number of lines in a code sample	Derived
Character_Count	Number of characters	Derived
Comment_Line_Count	Approximate count of lines beginning with common comment markers	Derived

Samples by origin and language (n)

Origin	C	C++	Java	Python	Total
AI	968	1221	1340	1471	5000
Human	769	1419	1605	1207	5000

problem_i	Sample_C	Generated	Language	Source
Prob3509		Human	Java	CodeNet
Prob292	import	Human	Java	CodeNet
Prob4407	from	AI	Python	ChatGPT-4
Prob4805	N,M =	Human	Python	CodeNet

Source: Dataset on Mendeley Data

Statistical methodology

1 Import the CSV | 2 Inspect the data | 3 Derive variables | 4 Summarize and plot | 5 Test hypotheses | 6 Interpret

Method we used:	It's purpose:
Mean, median, standard deviation	To summarize each code characteristic by origin
Boxplots and histogram	Used to show distributions, skew and outliers.
Welch's two-sample t-test	We used it to compare group means without assuming equal variances
95% confidence interval	Took as a plausible range for the difference in means
Chi-square test of independence	To test association between origin and language

Welch t-tests (line count, character count)

H_o: μ(AI) = μ(Human)

H_1: μ(AI) ≠ μ(Human)

Chi-square test (origin vs language)

H_o: origin and language are independent

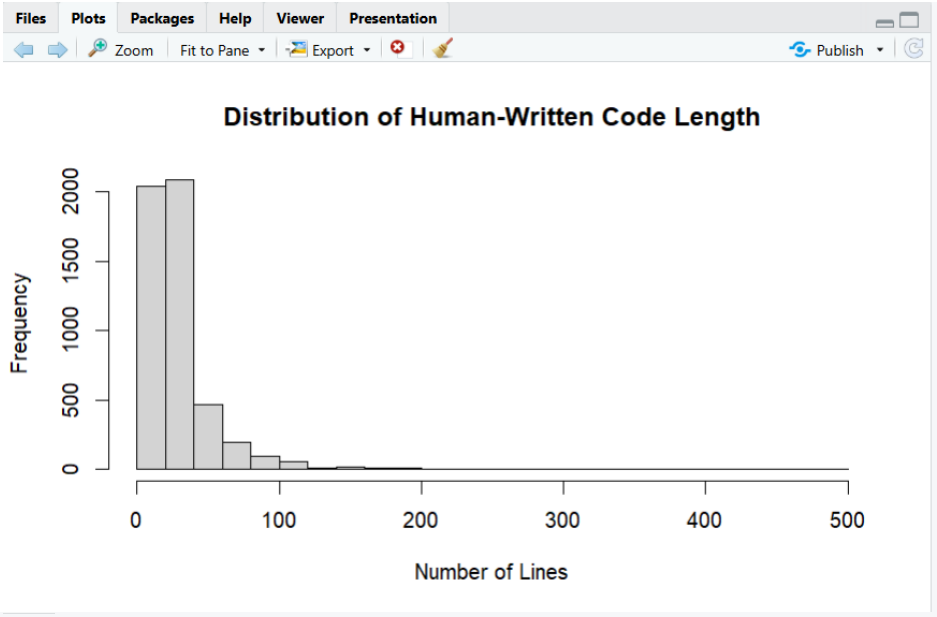
H_1: origin and language are associated

Significance level α = 0.05 (95% confidence). We approximated the comment counts (lines starting with # or /* or //).

R code implementation

```
> data$Character_Count <- nchar(data$Sample_Code)
> head(data[, c(
+   "Generated", "Language",
+   "Line_Count", "Character_Count"
+ )])
  Generated Language Line_Count Character_Count
1     Human     Java         24             473
2     Human     Java         30             444
3        AI   Python         15             380
4     Human   Python         19             396
5     Human     Java         88            1491
6     Human     Java         19             420
> aggregate(
+   Line_Count ~ Generated,
+   data = data,
+   FUN = function(x) c(
+     Mean = mean(x, na.rm = TRUE),
+     Median = median(x, na.rm = TRUE),
+     SD = sd(x, na.rm = TRUE),
+     Minimum = min(x, na.rm = TRUE),
+     Maximum = max(x, na.rm = TRUE)
+   )
+ )
  Generated Line_Count.Mean Line_Count.Median Line_Count.SD Line_Count.Minimum
1        AI        25.53880          21.00000      18.70779             2.00000
2     Human        29.58920          23.00000      25.53546             1.00000
  Line_Count.Maximum
1          230.00000
2          499.00000
```

```
> # Summarize the comment-line count by code origin
> aggregate(
+   Comment_Line_Count ~ Generated,
+   data = data,
+   FUN = function(x) c(
+     Mean = mean(x),
+     Median = median(x),
+     SD = sd(x)
+   )
+ )
  Generated Comment_Line_Count.Mean Comment_Line_Count.Median Comment_Line_Count.SD
1        AI                1.707200                 1.000000              1.874506
2     Human                2.397400                 1.000000              5.594187
> # Summarize comment-line counts by origin and language
> aggregate(
+   Comment_Line_Count ~ Generated + Language,
+   data = data,
+   FUN = function(x) c(
+     Mean = mean(x),
+     Median = median(x)
+   )
+ )
  Generated Language Comment_Line_Count.Mean Comment_Line_Count.Median
1        AI        C                2.6590909                 2.0000000
2     Human        C                1.7867360                 1.0000000
3        AI      C++                2.8402948                 2.0000000
4     Human      C++                6.4108527                 3.0000000
5        AI     Java                0.7671642                 0.0000000
6     Human     Java                0.5052960                 0.0000000
7        AI   Python                0.9966010                 0.0000000
8     Human   Python                0.5840928                 0.0000000
```




```
        Welch Two Sample t-test

data:  Line_Count by Generated
t = -9.0477, df = 9165.1, p-value < 2.2e-16
alternative hypothesis: true difference in means between group AI and group Human is not equal to 0
95 percent confidence interval:
 -4.927932 -3.172868
sample estimates:
   mean in group AI mean in group Human
            25.5388             29.5892

> # Test for a difference in mean character counts
> character_test <- t.test(
+   Character_Count ~ Generated,
+   data = data
+ )
> character_test

        Welch Two Sample t-test

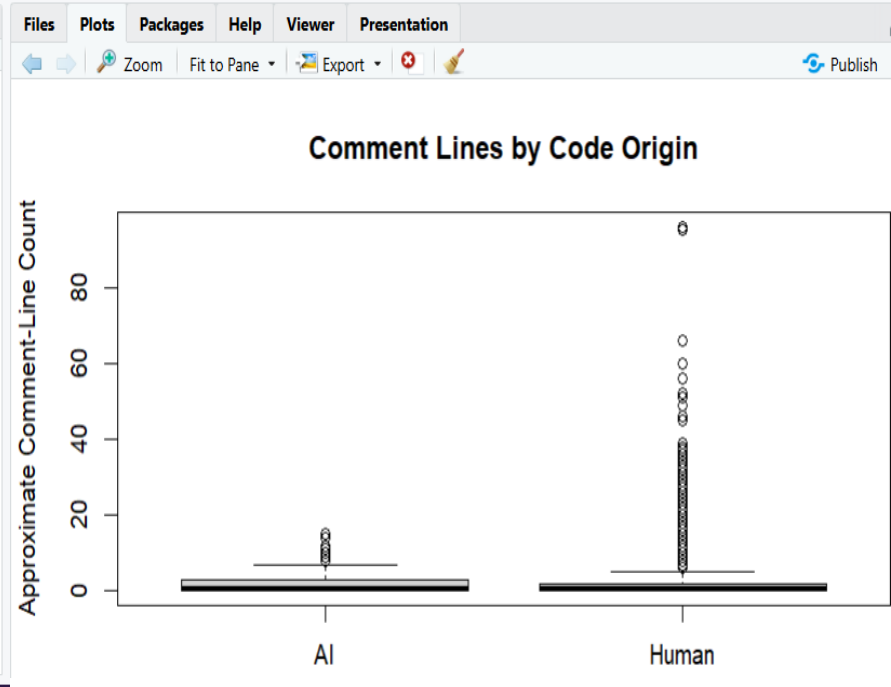
data:  Character_Count by Generated
t = 10.876, df = 9986.6, p-value < 2.2e-16
alternative hypothesis: true difference in means between group AI and group Human is not equal to 0
95 percent confidence interval:
  99.00083 142.53077
sample estimates:
   mean in group AI mean in group Human
           678.6266            557.8608

> # Test association between origin and language
> language_test <- chisq.test(
+   table(data$Generated, data$Language)
+ )
> language_test

        Pearson's Chi-squared test

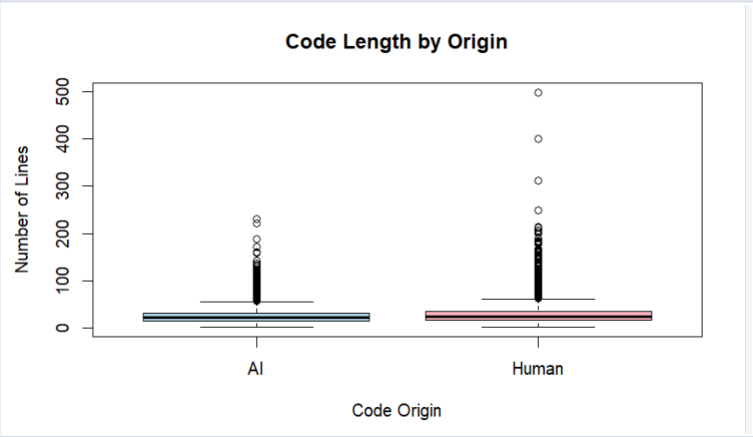
data:  table(data$Generated, data$Language)
X-squared = 87.519, df = 3, p-value < 2.2e-16
```



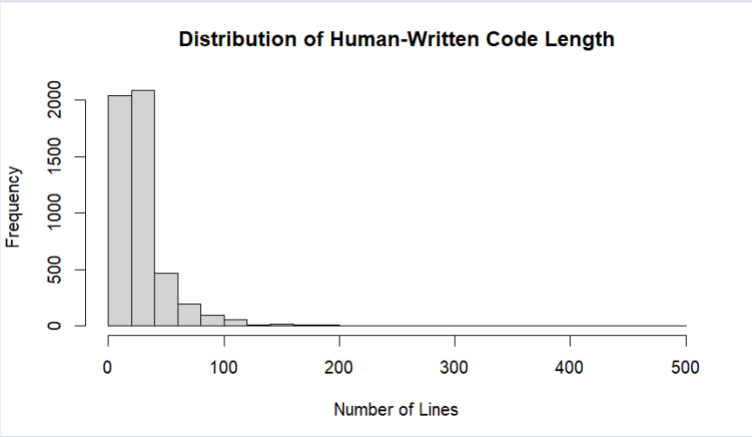


Preliminary descriptive results

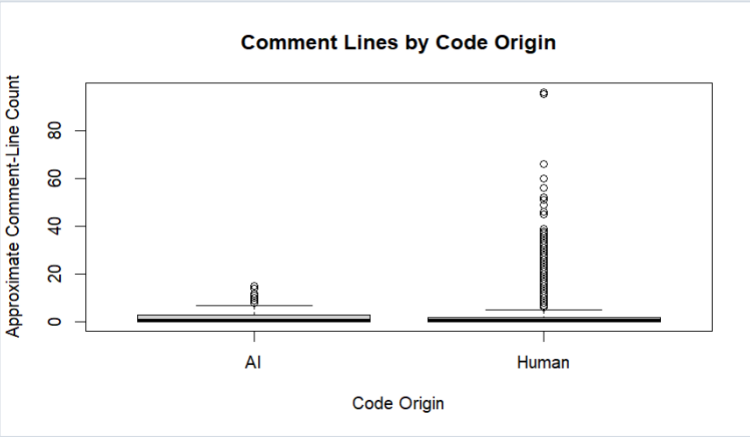
Measure	AI mean	Human mean	AI median	Human median	AI SD	Human SD
Lines per sample	25.54	29.59	21	23	18.71	25.54
Characters per sample	678.63	557.86	527	417.5	545.73	564.46
Approx. comment lines	1.71	2.40	1	1	1.87	5.59



Code length: medians are close yes, 21 vs 23 and both the groups have long right tails (basically more code lengths are shorter with a few high number).



Human code length is strongly right skewed means a few very long programs pull the mean up.



Human comment counts have a long tail we see maximum 96 vs 15 for AI.

By language, AI has the higher mean comment line count in C, java and python, human in C++ (6.41 vs 2.84). Also, # also matches C/C++ #include and #define lines, so C/C++ counts are inflated. These findings are descriptive, for this dataset only. In future, we may come up with a way to solve this issue.

Hypothesis test results

Test	Estimate (AI /human)	95% CI	Test statistic	p-val	decision
Mean line count (Welch t-test)	−4.05 lines	[−4.93, −3.17]	t = −9.05, df = 9165.1	$< 2.2 \times 10^{-16}$	**Reject H_0**
Mean character count (Welch t-test)	+120.77 characters	[99.00, 142.53]	t = 10.88, df = 9986.6	$< 2.2 \times 10^{-16}$	**Reject H_0**
Origin vs language (chi-square)	-	-	$\chi^2 = 87.52$, df = 3	$< 2.2 \times 10^{-16}$	**Reject H_0**

Line count

We see that human samples, average about 4 more lines than AI samples (29.59 vs 25.54)

Character count

Here, AI samples average about 121 more characters despite having more fewer lines. The cause (formatting, comments, structure maybe) is not yet tested yet.

Origin vs language

The language mix differs by origin (python is 29% of AI samples vs 24% of human), so within language comparisons are needed. We will explore this in the next review 2.

p values are tiny partly because n = 10000. Effects are small (Cohen's d ≈ 0.18 and 0.22; Cramér's V ≈ 0.09). Independence by problem ID is not yet checked.

Limitations

Balanced by design

We only have about 5000 samples per group, so the data say nothing about how common AI code is in the real world.

Different sources

Human code is from CodeNet, AI code from ChatGPT different models, differences may reflect collection, not authorship alone.

Unequal language mix

Origin and language are associated, so overall comparisons mix in language effects. We may explore this in later reviews (planned for review 2).

Possible dependence

6849 unique problem ids for 10000 samples. Also samples on the same problem may not be independent. We'll investigate it too.

Simple measures

Counts of lines, characters and comments do not measure correctness, logic or quality.

Approximate comment count

Counts lines starting with #, // or /*. The # also matches C/C++ #include and #define lines and block and inline comments maybe missed.

Work we completed and our next steps:

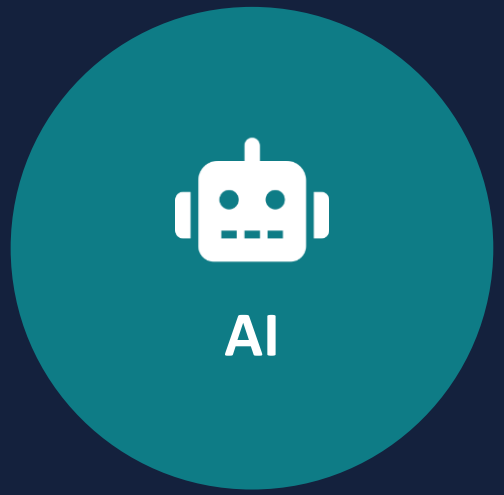
Completed for Review 1

- Dataset downloaded, imported in R and we checked it (10000 rows, 5 columns, no missing code, no duplicate rows)
- Derived variables: line, character and approximate comment-line counts
- Descriptive statistics and plots by code origin, all pasted
- Welch t-tests, 95% confidence intervals and chi-square test
- Literature review

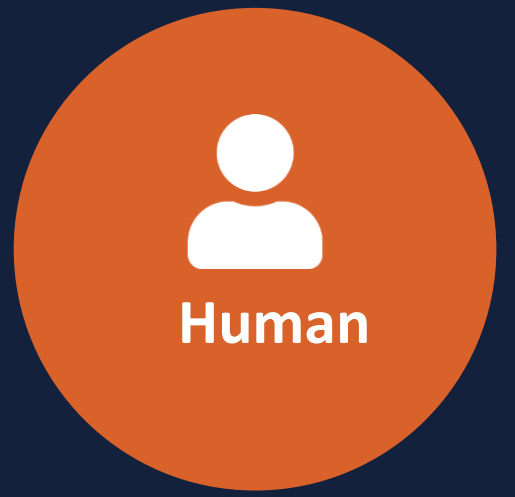
Planned for Reviews 2 and 3

- We will repeat the comparisons within each of the four languages to investigate deeper
- Test comment-line counts formally
- Check independence (problem ids) and robustness to skew (Wilcoxon rank sum test)
- Finalize interpretation, report and conclusions

Conclusion so far: measurable code characteristics differ between the two groups in this dataset, warranting further analysis by language and task.



Thank You!



Dataset: Boukili, G., El Garouani, S., & Riffi, J. (2026). A dataset for human-written and AI-generated code source classification. Data in Brief, 65, 112527. https://doi.org/10.1016/j.dib.2026.112527